OPERATING SYSTEMS – ASSIGNMENT 1

TASK 1 – Evaluate the Requirements of Each Computing Environment (4 Marks)
CO1 | Bloom's Level: K5 – Evaluate

Introduction
A smart hospital uses different kinds of computers for patient monitoring, communication, medical imaging and patient records. Each environment has different operating-system needs. Therefore, using one general-purpose operating system for every device may not provide the best response time, reliability, battery life and security. The requirements should be ranked according to their importance to patient care.

1. Ranked requirements for each environment

Environment	Priority ranking (highest to lowest)	Justification
Bedside patient monitors	**1.** Response time **2.** Reliability **3.** Security **4.** Throughput **5.** Battery	The monitor must detect an abnormal reading and raise an alarm within 10 milliseconds. A missed or delayed alarm can affect patient safety. It should run continuously and protect patient data. It handles a limited amount of sensor data, so high throughput is less important. If it is mains-powered, battery life is a lower priority.
Nurses' handheld tablets	**1.** Battery life **2.** Security **3.** Reliability **4.** Response time **5.** Throughput	Tablets must last through a working shift and protect patient records. They should remain reliable while nurses access records and communicate. A quick user interface is useful, but a strict 10 ms deadline is not normally required. Their workload is moderate.
Doctors' desktop workstations	**1.** Security **2.** Reliability **3.** Response time **4.** Throughput **5.** Battery	Workstations access confidential records and clinical applications, so access control is essential. They must be dependable and responsive for daily work. They may handle several applications, but they are normally connected to mains power, making battery life unimportant.
Central patient-record database	**1.** Reliability / availability **2.** Security **3.** Throughput **4.** Response time **5.** Battery	The database must remain available because doctors and nurses depend on patient records. Strong authentication, permissions and data protection are essential. It must support many users and transactions. Fast queries matter, but continuous availability and correctness are more critical. Battery life is not relevant.
MRI imaging server	**1.** Throughput / processing power **2.** Reliability **3.** Response time **4.** Security **5.** Battery	MRI scans involve large amounts of data and computational work. The server needs multiple processors, sufficient memory and high data-transfer capacity. It must process scans reliably and provide results within a useful time. It is mains-powered, so battery life is not a concern.
Partner-hospital link	**1.** Security **2.** Reliability **3.** Response time **4.** Throughput **5.** Battery	Shared records must be transferred securely between hospitals. The connection should be dependable and protect data in transit. Response time and bandwidth affect how quickly records are shared, but patient confidentiality and integrity come first.

2. Conflicting requirements

Some hospital requirements conflict with one another:

• **Fast response vs. many users:** Patient monitors need predictable alarm response. A general-purpose system busy serving many tasks may delay an alarm.
• **Battery life vs. performance:** Tablets need low power consumption, while high processing speed and continuous wireless communication can use more battery.
• **Security vs. convenience:** Strong authentication and access checks protect records, but they can add steps or small delays for users.
• **High throughput vs. reliability:** The imaging server must process large scan files quickly, but it must also avoid failures and protect data.
• **Availability vs. maintenance:** Database updates and maintenance must be planned so that patient-record access is

not interrupted.

3. Requirements critical to patient safety

The most safety-critical requirement is the bedside monitor's guaranteed alarm response within 10 milliseconds. Reliability is also essential because the monitor must operate continuously. The central database must be highly available and protect the correctness and confidentiality of patient records. The imaging server must process scans reliably so that clinical decisions are based on correct images. Security is important across all environments because incorrect or unauthorised access to medical information can harm patients.

4. Lessons from the history of operating systems

• **Batch systems:** Early batch systems processed jobs in groups without continuous user interaction. This idea is suitable for non-urgent hospital work such as scheduled backups and bulk report generation, but not for real-time patient alarms.
• **Time-sharing systems:** Time-sharing allows many users to share a computer interactively. It is useful for doctors' workstations and the central database, where many staff members access services.
• **Personal-computer systems:** Desktop operating systems provide interactive applications and user-friendly interfaces. They suit doctors' desktop workstations.
• **Mobile systems:** Mobile operating systems manage wireless communication, touch interfaces and power use. They suit nurses' handheld tablets because battery life and mobility are important.

5. Alternative concept: one general-purpose OS for every device

If the whole hospital used one general-purpose operating system, administration and software support could become simpler. Staff would have a more consistent environment, and IT teams could use similar update and monitoring procedures. However, this approach is not suitable for every device because their needs are different.

Worst-case alarm response time: A general-purpose OS usually does not guarantee a fixed maximum scheduling delay. If the CPU is busy with other work, an alarm task may wait for an unpredictable time. Therefore, its worst-case delay may be unbounded for a strict deadline analysis. A real-time operating system is designed to provide predictable scheduling and can be configured and verified to meet the 10 ms deadline. A deadline must still be validated on the actual hardware and workload.

Memory required: A full general-purpose OS usually needs more memory and storage for its services, drivers, user interface and background processes. This may be unsuitable for small bedside monitors. A compact embedded or real-time OS can use fewer resources. Tablets and workstations can support larger operating systems because they have more memory and storage.

Administration effort: Using one OS may reduce the number of platforms that IT staff must learn and maintain. However, the same OS may need different configurations on every device, and unnecessary services can increase security and resource overhead. Separate, purpose-selected operating systems require more specialised administration but better match each environment.

6. Final evaluation

The hospital should not use one general-purpose OS on every device. A suitable plan is to use a real-time or embedded OS for bedside monitors, a mobile OS for nurses' tablets, a desktop OS for doctors' workstations, and a reliable server OS for the database and MRI imaging systems. Secure communication should connect the partner hospitals. This mixed approach gives priority to patient safety, availability, security, battery life and processing performance according to each environment.

Conclusion: Different hospital environments have different operating-system requirements. A specialised OS for each environment is safer and more efficient than forcing every device to use one general-purpose OS.

Note: This answer is prepared from the Task 1 requirements in the uploaded CSA04 Operating Systems Assignment 1 document. Add your name, register number and any evidence required by your faculty before submission.